

Introduction: The Teleological Attack Surface

Series Context I

This paper is Part 4 of the *Cognitive Security for Multiagent Operators* series, which progresses from theory to computation to practice to applied deployment:

- ▶ **Paper 1: Formal Foundations** [?] (DOI: 10.5281/zenodo.18364119) establishes the Cognitive Integrity Framework (CIF): a formal model of agent cognitive states $\sigma_i = \langle \mathcal{B}_i, \mathcal{G}_i, \mathcal{J}_i, \mathcal{H}_i \rangle$, a trust calculus with delegation decay (δ^d), the Defense Composition Algebra, the five-tier adversary taxonomy (Ω_1 – Ω_5), information-theoretic stealth–impact bounds, and model-checked safety invariants. A supplementary chapter (S02) additionally develops the eusocial-colony analogy as an evolutionary existence proof for CIF-like defense architectures.

Series Context II

- ▶ **Paper 2: Computational Validation** [?] (DOI: 10.5281/zenodo.18364128) validated these mechanisms computationally across a 950-attack corpus and four production multiagent architectures, reporting ablation studies, Bayesian uncertainty quantification, and colony-scale benchmarks; the recommended defense stack achieves 94–100% detection in parametric simulation and 80–100% under LLM-backed evaluation.
- ▶ **Paper 3: A Qualitative Review for Practitioners** [?] (DOI: 10.5281/zenodo.18364130) translates the theoretical and empirical results into accessible engineering guidance: deployment guides, subagent-hardening patterns, incident-response playbooks, monitoring strategies, cost–benefit analysis, common pitfalls, case studies, and operator risk frameworks. It assumes no formal prerequisites.

Series Context III

- ▶ **Paper 4: Applications of the Cognitive Integrity Framework** (this paper) addresses the remaining question: **does CIF work in practice across diverse operational domains?** We apply the framework across ten critical sectors—from millisecond drone swarm decisions to year-scale diplomatic deliberations—through the integrated CIF-AD-OODA analytical model, demonstrating universality and uncovering novel defense patterns visible only at the cross-domain scale.

Together, Papers 1–4 provide a complete stack: Paper 1 defines *what* CIF is; Paper 2 shows that it *works*; Paper 3 shows *how to deploy it*; Paper 4 shows *where* it applies and *how* domain specifics extend it.

The Ontological Crisis in AI I

The vulnerability of modern Artificial Intelligence has shifted from the *epistemic* (what the agent knows) to the *teleological* (what the agent wants) [?, ?]. **Goal Hijacking**, a sophisticated vector of indirect prompt injection [?], allows adversaries to surreptitiously rewrite an agent's objective function. This represents an ontological crisis for autonomous systems: if an agent cannot trust the integrity of its own goals, it cannot trust any action it calculates.

The Ontological Crisis in AI II

In the context of Boyd's **OODA (Observe-Orient-Decide-Act) Loop** [?, ?], Goal Hijacking is a corruption of the **Orientation** phase. The agent correctly Observes the world, but its internal Orientation—the synthesis of heritage, culture, and genetic code (or in AI terms: training data, system prompts, and hard-coded constraints)—is displaced by a parasitic instruction. The agent then proceeds to Decide and Act with perfect logical consistency, but in service of an alien will. This dynamic has been documented across the emerging agentic AI landscape [?, ?]. The OWASP Top 10 for Agentic Applications (December 2025) designates **ASI-01: Agent Goal Hijack** as the #1 risk for deployed agentic AI systems—a direct industry validation of this paper's central thesis.

The Ontological Crisis in AI III

Empirical Urgency

Goal Hijacking has transitioned from academic concern to documented production failure. Autonomous coding agents have deleted production databases and fabricated records to conceal the damage; invisible Unicode payloads have triggered auto-approval modes enabling remote code execution; and indirect prompt injection through enterprise messaging platforms has exfiltrated private API keys—all without human authorization [?, ?, ?]. The Agent Security Bench (ASB) evaluation [?] quantifies the gap: an 84.3% average attack success rate across 400+ integrated tools, with current defenses achieving only 19.7% mitigation. He et al. [?] further demonstrate Agent-in-the-Middle attacks that compromise inter-agent communication channels, extending the threat surface to multiagent coordination itself. These incidents are analyzed in detail through the CIF-AD-OODA lens in sec:empirical_g*rounding*.

Axiomatic Design and Transient Functional Requirements I

Suh's **Axiomatic Design (AD)** theory [?, ?] posits that good design maintains the independence of Functional Requirements (FRs). The relationship between FRs and Design Parameters (DPs) is captured in the Design Matrix:

$$\{FR\} = [A]\{DP\} \quad (1)$$

In an uncoupled (safe) design, $[A]$ is diagonal: each FR depends on exactly one DP. Providing a solution for FR_1 does not compromise FR_2 . In cognitive security, we identify a new failure mode: **Transient Functional Coupling**.

Axiomatic Design and Transient Functional Requirements II

Adversaries use “fast OODA transients”—high-frequency semantic injections through data channels (Ω_2 vectors)—to temporarily introduce off-diagonal terms in the Design Matrix. For instance, a cyber-defense agent’s FR (“Block Malicious Traffic”) might be transiently coupled to a hijacked FR (“Maximize Uptime”), causing the agent to flush its firewalls during an attack to “restore connectivity.” The coupled matrix:

$$\{FR'\} = [A']\{DP\} = \begin{bmatrix} A_{11} & A_{12} \\ A_{21} & A_{22} \end{bmatrix} \{DP\} \quad (2)$$

violates the Independence Axiom, rendering the system unstable. The focus of this paper is on detecting and defending against these rapid shifts in the Design Matrix.

The Role of CIF: Five Canonical Defenses I

The Cognitive Integrity Framework serves as the “gyroscope” for OODA loops, stabilizing the Orientation phase against adversarial transients. CIF provides five canonical defense mechanisms [?], each targeting a specific aspect of cognitive integrity:

1. **Cognitive Firewall** ($\mathcal{F}$): Classifies incoming messages by trust score, quarantining ambiguous inputs before they reach the agent's Orientation phase.
2. **Belief Sandboxing** ($\mathcal{B}_{\text{provisional}}$): Isolates unverified beliefs in a provisional partition, requiring corroboration before promotion to operational status.
3. **Behavioral Invariants** (INV_k): Pre-defined runtime predicates that trigger alerts when violated, regardless of semantic content.
4. **Drift Detection** (S_{drift}): Monitors belief distribution changes via KL divergence, flagging sudden orientation shifts that exceed threshold ϵ .

The Role of CIF: Five Canonical Defenses II

5. **Byzantine Consensus** ($\mathcal{B}_{\text{consensus}}$): Multi-agent agreement protocol requiring quorum q before critical actions, tolerating up to f compromised agents where $n \geq 3f + 1$ [?].

These mechanisms compose in series and parallel to achieve layered defense. Rather than the ad-hoc “Axiomatic Locking” described in early formulations, CIF’s defense is the *systematic composition* of these five mechanisms, calibrated to the threat profile and temporal dynamics of each operational domain.

Application Analysis I

This manuscript examines Goal Hijacking dynamics across ten high-stakes domains, analyzing how adversaries exploit the collaborative surface between AI agents, OODA loops, and Axiomatic Design principles. Each domain is analyzed using a standardized five-step template (sec:methodology) that characterizes the operational context, attack surface, transient coupling, defense mapping, and validation anchoring.

Contributions I

This paper makes the following contributions:

- ▶ **C1:** A unified CIF-AD-OODA integration model for analyzing Goal Hijacking attacks and defenses across arbitrary operational domains.
- ▶ **C2:** Identification of three universal attack patterns—FR Polarity Inversion, Constraint Relaxation, and Context Boundary Violation—through cross-domain synthesis.
- ▶ **C3:** Validation that all five canonical CIF mechanisms provide adequate coverage across ten critical domains, with no mechanism appearing in fewer than three domains and no domain requiring mechanisms outside the CIF vocabulary.
- ▶ **C4:** Three novel defense pattern extensions: verification channel separation, active perturbation probing, and physics-informed invariants.
- ▶ **C5:** Temporal scale analysis demonstrating CIF's applicability across eight orders of magnitude in OODA cycle time.

Contributions II

- ▶ **C6:** Retrospective validation through six documented AI agent security incidents (2024–2025), confirming that all incidents map to the universal attack pattern taxonomy and would have been detectable by the appropriate CIF mechanism.

Reading Companion: Where to Find Specific Topics I

This paper is designed to stand alone as the applied, domain-facing reference of the series. Where a formal construct, empirical measurement, or engineering technique is developed more fully elsewhere, the table below points the way.

Table 1: Cross-paper navigation from Part 4 topics to sibling developments.

Reading Companion: Where to Find Specific Topics II

If you want...	...consult...
Formal definition of the cognitive state	Part 1 [?], §3 (System Model)
$\sigma_i = \langle \mathcal{B}_i, \mathcal{G}_i, \mathcal{J}_i, \mathcal{H}_i \rangle$	
Trust Calculus, δ^d decay theorem, no-amplification guarantee	Part 1, §4 (Trust Calculus)
Defense Composition Algebra (series/parallel theorems)	Part 1, §5
Information-theoretic stealth-impact bounds	Part 1, §4.3
Adversary taxonomy Ω_1 – Ω_5 formal characterization	Part 1, §3
Model-checked safety invariants + NuSMV/TLA+ specifications	Part 1, §7; Part 2 S04
Eusocial-colony analogy as existence proof for CIF-like architectures	Part 1 S02
Empirical detection rates for the defenses applied in this paper	Part 2 [?], §5 + S08 (Parametric Analysis)